

Long-term Comparison of Semitendinosus and Gracilis Tendon Versus Patellar Tendon Autografts for Anterior Cruciate Ligament Reconstruction

A 17-Year Follow-up of a Randomized Controlled Trial

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Background: Short-term and mid-term differences between hamstring and patellar tendon autografts for anterior cruciate ligament (ACL) reconstruction are well documented. Systematic reviews highlight the lack of long-term results between the two grafts.

Hypothesis: Seventeen years after ACL reconstruction, no difference will be found in functional outcome, quality of life, and graft failure between patients with patellar tendon (PT) or semitendinosus and gracilis tendon (STG) autografts; however, a significant difference will be seen in the prevalence of osteoarthritis.

Study Design: Randomized controlled trial; Level of evidence, 2.

Methods: Sixty-four patients were included in this prospective study (32 in each group). A single surgeon performed primary ACL reconstruction in alternating sequence. Forty-eight patients (24 in each group) were evaluated 17 years after ACL reconstruction: A clinical assessment was made based on the International Knee Documentation Committee (IKDC) form, instrumented laxity was measured with KT-1000 arthrometer, and radiography of the operated knee was conducted and assessed for degenerative disease. The Lysholm questionnaire and the Short Form-36 version 2 questionnaire were filled out by the patients.

Results: At the 17-year follow-up, no statistically significant differences were seen with respect to graft failure (2 reruptures in the semitendinosus and gracilis tendon [STG] group [6.3%] and 3 reruptures in the PT group [9.4%]) and functional outcome. Increased instrumented laxity (>3 mm) measured with KT-1000 arthrometer was seen in significantly more patients in the STG group (8 in the STG group compared with 2 in the PT group; $P = .03$) with a mean side-to-side difference of 2.17 ± 1.86 mm in the STG group compared with 1.33 ± 1.93 mm in the PT group. A significant difference was found in frequency of knee osteoarthritis (OA)—100% in the PT group compared with 71% in the STG group ($P = .004$). Patients in the PT group tended to have higher grade OA according to the IKDC grading system, with an average grade of 1.46 in the PT group compared with 1 in the STG group ($P = .055$). The degenerative changes in the PT group were more common in the medial and patellofemoral compartments ($P = .003$ and $P = .04$, respectively).

Conclusion: Both autografts provided good to excellent subjective outcomes. No significant differences were noted in graft failure and clinical instability. However, significantly more patients in the STG group had increased instrumented anteroposterior translation measured with KT-1000 arthrometer, and there was a greater prevalence of knee OA at 17 years after surgery in the PT group.

Keywords: anterior cruciate ligament (ACL) reconstruction; hamstring tendon; patellar tendon; ACL and osteoarthritis; quality of life; long-term; randomized controlled trial

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Anterior cruciate ligament (ACL) reconstruction is one of the most common orthopaedic techniques performed worldwide. The incidence of ACL injury was between 40 and 60 in 100,000 in 2014 and is increasing every year.^{18,24} Next to symptomatic instability, the main problems attributable to ACL injury are the long-term consequences. Incidence of knee osteoarthritis (OA) increases to 15% to 20% in the ACL-deficient knee, which represents a 10-fold increase. More than 50% of patients who have ACL injury will

develop symptomatic OA in the following 10 to 20 years.^{11,17,24} The main goals of ACL reconstruction are stabilization of the knee joint and therefore prevention of additional chondral and meniscal damage, restoration of normal knee kinematics, and prevention of degenerative changes. However, increased incidence of OA has been found even after ACL reconstruction.^{22,27-29} The choice of graft material for ACL reconstruction is still controversial. Numerous studies have compared various types of ACL grafts (autografts, allografts, and synthetic grafts). Due to numerous biological advantages, the most commonly used grafts for ACL reconstruction are the hamstring tendon (HT) autografts and patellar tendon (PT) autografts. The advantages of the HT graft are the lower donor site morbidity and reduced anterior knee pain and kneeling pain.^{9,21,37} The only major disadvantage of HT graft is more frequent tunnel widening.^{6,24} In contrast, the advantages of PT graft are lower retear rate compared with HT graft, earlier ligamentization and vascularization, and therefore earlier return to activity.^{7,9,19,24,33} The disadvantages of PT graft are higher donor site morbidity and increased incidence of OA.^{24,27,32}

Two systematic reviews of randomized controlled trials (RCTs)^{16,25} comparing HT and PT autografts showed more anterior knee pain, kneeling pain, and extension loss and higher rates of OA with the use of PT graft; the reviews found more knee laxity resulting in positive pivot-shift test with the use of HT graft. Similar results were published in a Cochrane review by Mohtadi et al.²¹ The review also highlighted the lack of high-quality long-term results, with studies reporting outcomes with follow-up ranging from 7 to 14 years. We found only one prospective comparative study with a follow-up of 20 years.³² The present study is a follow-up of previously reported 5-year and 11-year follow-up studies,^{27,28} which showed that patients with PT grafts had a greater prevalence of OA and that the PT group had a significant difference in positive pivot shift. The hypothesis of this 17-year follow-up study is that patients with PT versus HT autografts will have no differences in functional outcome, quality of life, and graft failure; however, patients in these two groups will have a significant difference in the prevalence of OA.

METHODS

Patient Selection

In this study, the key indication for surgical reconstruction was a clinical diagnosis of ACL rupture in a patient desiring to return to his or her preinjury level of activity. Patients with associated ligament injury, previous meniscectomy, an abnormality seen on radiograph, or an abnormal contralateral knee joint or who did not wish to participate in a research program were excluded from the study. From June 1999 to March 2000, 64 patients underwent ACL reconstruction and were enrolled in the study. During that time, the surgeon performed 75 ligament reconstructions, which means that 85% of patients were included in the study. Patients underwent a primary

arthroscopically assisted ACL reconstruction with graft randomization into a PT or HT group according to operative registration list position (even number = PT, odd number = HT). In 32 patients, ACL reconstruction was performed with doubled semitendinosus and gracilis tendon autograft (STG group), while in the other 32 patients, ACL reconstruction was performed with PT autograft (PT group). All patients signed an informed consent form. The study was approved by the National Ethical Committee. Patients who underwent ACL reconstruction on the contralateral knee or who required revision surgery during the follow-up period were excluded from the analysis.

Operative Technique

All procedures were performed by the first author (M.S.). Apart from the graft harvesting, the surgical technique was identical. All patients were examined under anesthesia (Lachman, drawer, pivot-shift, and rotation tests were documented), after which routine diagnostic arthroscopy and meniscal surgery were performed, followed by the ACL reconstruction. For the patients randomized to the PT group, a graft between 9 and 10 mm in diameter was harvested via longitudinal incision as a free autograft. The defects of the patella and the proximal tibia were not bone grafted. For the patients in the STG group, the graft was harvested through a longitudinal incision at the site of the pes anserinus insertion. The sartorius fascia was split, and the gracilis and semitendinosus tendons were harvested with a tendon stripper. The tendons were cleaned and looped over a No. 2 Vicryl absorbable suture to create a quadruple graft, and the graft was between 7 and 9 mm in diameter. The graft was manually pretensioned up to 20 pounds. The femoral tunnel was drilled through the anteromedial portal before tibial tunnel drilling. The femoral tunnel was drilled through the center of the femoral anatomic footprint, between the anteromedial and posterolateral bundles, while the tibial tunnel was drilled through the central part of the anatomic footprint. Drill guides were used to confirm the correct position of the tunnels. To drill the femoral entry point, the surgeon used the "bull's eye" drill guide (Linvatec). Tunnel size in the PT group was made 1 mm larger than the bone block size. In the STG group, tunnel size was matched to the cross-sectional size of the graft. A marking suture using No. 0 absorbable suture was set 2.5 cm from the femoral end of the graft to ensure good entry of the graft into the tunnel and to prevent the graft from twisting around the screw during insertion. The graft was inserted via the tibial tunnel into a blind femoral tunnel. A left thread or right thread roundhead, cannulated titanium interference screw (RCI; ART-MAM) was used for femoral graft fixation.²⁸ In right knees a screw with a left thread was used, and in left knees a screw with a right thread was used. In the right knee, the left thread screw turns in the left direction and pushes the graft away from the lateral wall of the intercondylar notch, and vice versa. This process prevents graft impingement against the lateral wall of the intercondylar notch. Tibial anatomic joint line fixation was achieved

with a bioabsorbable interference screw (Linvatec) in an outside-in direction at a knee flexion angle of approximately 10° and manual pretensioning of the graft.

Rehabilitation

All patients underwent rehabilitation according to the same accelerated protocol, which permitted immediate full weightbearing and full range of motion.³⁰ A rehabilitation brace was used for 3 weeks postoperatively. The splint was not worn during rehabilitation or during the night. The importance of reaching full extension was emphasized from the beginning. The patients had to obtain an arc of at least 0° to 90° before they were discharged from the hospital. Closed kinetic-chain exercises were started immediately after the operation. The leg extension exercise against resistance in the arc from 45° to 0° was not allowed during the initial 6 postoperative weeks. Running was allowed at 8 weeks and contact sports at 6 months under 4 conditions: (1) no effusion, (2) full range of motion, (3) stable knee, and (4) obtained muscle strength of 90% compared with the contralateral side.

Follow-up Evaluation

Clinical evaluations of knee function and stability and radiologic evaluations were performed preoperatively; at 2 weeks, 6 weeks, 3 months, and 5 months after surgery; and at 5-year, 11-year, and 17-year follow-up examinations. Clinical evaluations at 17-year follow-up were performed by the second author (D.S.). The physical therapist (K.S.) performed all arthrometric assessments. At 11-year and 17-year follow-up, the Lysholm knee score questionnaire³¹ and Short Form-36 version 2 questionnaire (Standard Form)²⁰ were sent to the patients by mail and completed in their homes. The International Knee Documentation Committee (IKDC) activity level was used to assess the functional outcome.¹² Patients were evaluated by use of the IKDC score according to the 2000 IKDC Knee Examination Form. Clinical ligament testing was performed by means of the Lachman test and a pivot-shift test; side-to-side differences were recorded. Objective anteroposterior (AP) knee stability was determined by using the KT-1000 arthrometer (MEDmetric) at manual maximum tension and a knee flexion angle of 25°. We selected 3 mm as the upper limit of normality of side-to-side difference in anterior tibial displacement. After the evaluation of the IKDC score, we graded these parameters as A (normal), B (nearly normal), C (abnormal), and D (severely abnormal) compared with the patients' contralateral knee. The patients were classified as having subjective anterior knee pain if they registered pain with any of the following activities: (a) during or after activity, (b) during stair-climbing, or (c) during squatting or kneeling. Before surgery and at 5 years, 11 years, and 17 years after surgery, 30° of posteroanterior flexion weightbearing and weightbearing lateral radiographs were obtained. Patients with abnormal radiographs before surgery were not included in the study. Radiologic

evaluation was conducted according to IKDC recommendations (A, normal; B, minimal changes and barely detectable joint-space narrowing; C, minimal changes and joint-space narrowing up to 50%; D, more than 50% joint-space narrowing). The single-legged hop test was used to evaluate functional performance.

Statistical Methods

In April 1999, a hospital database with all preoperative and postoperative variables was established. During the follow-up period, we updated the database and performed statistical analysis of variables. For continuous numerical variables, mean (SD) values are presented. The unpaired *t* test was used for the normally distributed numerical data. A nonparametric analog (Mann-Whitney test) was used where appropriate. Paired *t* test was used to compare preoperative and postoperative values (preoperative Lysholm score vs postoperative Lysholm score). The chi-square test was used to compare categorical variables (overall IKDC scores, the severity of OA). The correlation between chondrolysis or subtotal meniscectomy and OA was calculated with univariate logistic regression. Graft survival was calculated and compared between the two groups by use of the Kaplan-Meier survival method. Comparisons of survival curves were made with log-rank tests and univariate Cox regression. A *P* value less than .05 was considered statistically significant.

RESULTS

Of the 64 patients in the study, 48 (75%) were analyzed at the 17-year follow-up (Figure 1). Of the 16 patients not analyzed, 5 were not available, and the remaining 11 patients were excluded from the study due to rupture of the ACL of the contralateral knee or revision ACL surgery. Three patients from the STG group as well as 3 patients from the PT group underwent an ACL reconstruction of the contralateral knee. Furthermore, 2 patients (6.3%) from the STG group and 3 patients (9.4%) from the PT group had a rerupture of the ACL and underwent a revision ACL reconstruction at our hospital. A fourth patient with a possible rerupture in the PT group reported in the prior 11-year follow-up study had a revision surgery in another hospital, where clinicians found only a partial tear in the ACL graft and did not perform a repeat reconstruction. The patient was invited to the 17-year follow-up but was unavailable. No new graft ruptures occurred between the 11-year and 17-year follow-ups.

A total of 48 patients (24 in each group) were clinically examined and radiologically evaluated at 17-year follow-up. Graft survival in both groups can be seen in the Kaplan-Meier survival curve (Figure 2). No significant differences were found in graft rupture between the two groups (*P* = .639).

No significant difference was found between the two groups regarding demographic, preoperative, and intraoperative parameters. The time to surgery was not

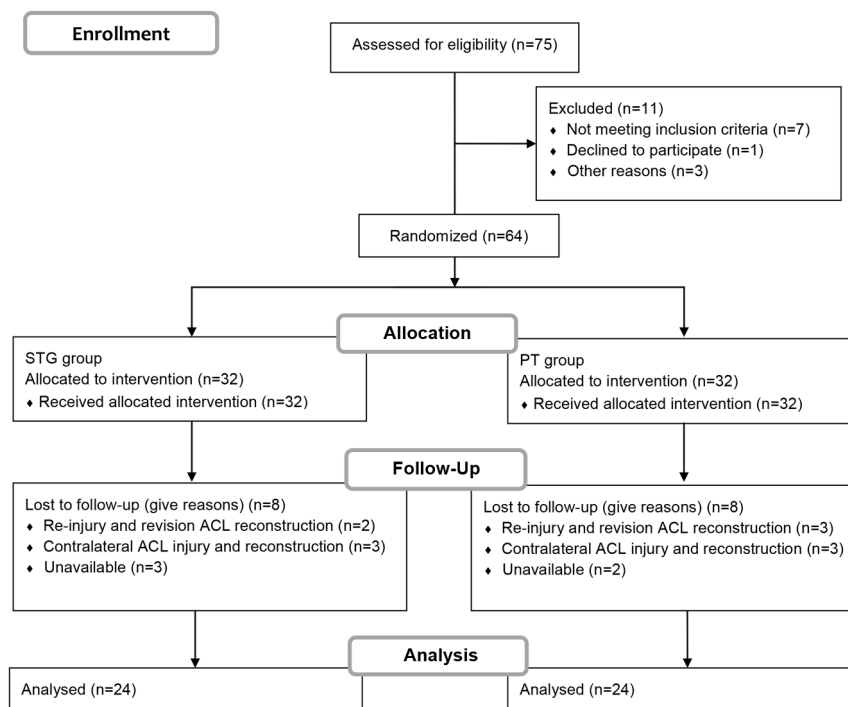


Figure 1. Patient sampling and selection. ACL, anterior cruciate ligament; PT, patellar tendon; STG, semitendinosus and gracilis tendon.

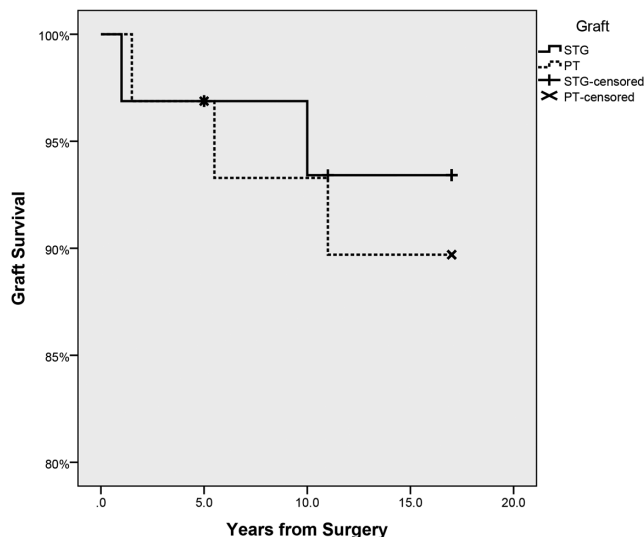


Figure 2. Cumulative survival of anterior cruciate ligament graft compared between the two groups. PT, patellar tendon; STG, semitendinosus and gracilis tendon.

significantly different between the two groups. Acute reconstruction (<12 weeks) was performed in 4 patients (17%) in each group. Meniscectomies were performed in 15 patients (63%) in the STG group and 18 patients (75%) in the PT group. No significant difference was found between the groups regarding partial, subtotal meniscectomies or meniscal repair.

Arthroscopic meniscal repair was performed in 2 patients in the STG group (8%) and 1 patient in the PT group (4%). All procedures were performed with nonabsorbable sutures with an inside-out technique. We found chondrolysis (Outerbridge grade II) at the time of surgery in 3 patients in each group (13%) (Table 1).

From the 48 patients analyzed at the 17-year follow-up, 2 patients from the STG group had an arthroscopic procedure in which a full-thickness chondral lesion was found and treated with microfracture (12 and 14 years after ACL reconstruction). One patient from the PT group underwent open-wedge valgization osteotomy of the tibia due to grade III degenerative joint disease findings in the medial compartment (16 years after ACL reconstruction). One patient had surgery on the contralateral knee (meniscal surgery).

Preoperative versus postoperative Lysholm knee scores showed significant improvement in both groups (STG, $P < .001$; PT, $P < .001$). At the 17-year follow-up, no significant differences were found in Lysholm score between the two groups; the mean Lysholm score was 94 (SD, 9.4) in the STG group and 93 (SD, 8.2) in the PT group, with 14 (58.3%) and 13 (56.5%) patients still achieving an excellent score of more than 95, respectively (Table 2). The average Lysholm scores during the follow-up period are shown in Figure 3.

Fifteen patients (63%) in the STG group and 10 patients (46%) in the PT group were still at their preinjury level of activity ($P = .246$). We also found no difference between the two groups in Tegner activity score, with 7 patients (29%) in the STG group and 4 patients (17%) in the PT group still participating in competitive sports (Tegner score ≥ 7) ($P = .303$). The single-legged hop test was used as a functional

TABLE 1
Comparison of Randomized Groups^a

	STG (n = 24)	PT (n = 24)	P Value
Age at follow-up, y, mean $\pm$ SD	42.5 $\pm$ 7.5	45.5 $\pm$ 8.7	.347
Sex			.56
Male	13 (54)	15 (63)	
Female	11 (46)	9 (37)	
Body mass index, mean $\pm$ SD	24.5 $\pm$ 3.1	24.0 $\pm$ 3.5	.722
IKDC preinjury activity level 1	17 (71)	18 (75)	.75
Tegner preinjury score >6	16 (67)	19 (79)	.33
Mean preoperative Lysholm score, mean $\pm$ SD	61.4 $\pm$ 25.9	56.6 $\pm$ 20.6	.614
Mean time to surgery, mo, mean $\pm$ SD	27.5 $\pm$ 43.5	22.1 $\pm$ 28.8	.616
Acute reconstruction	4 (17)	4 (17)	$\geq .999$
Chronic reconstruction	20 (83)	20 (83)	$\geq .999$
Meniscal surgery	15 (63)	18 (75)	.63
Meniscal repair	2 (8)	1 (4)	.55
Partial meniscectomy	6 (25)	10 (42)	.22
Subtotal meniscectomy	9 (38)	8 (33)	.76
Chondrolysis (Outerbridge grade II) at time of surgery	3 (13)	3 (13)	$\geq .999$

^aData are provided as n (%) unless otherwise noted. IKDC, International Knee Documentation Committee; PT, patellar tendon; STG, semitendinosus and gracilis tendon.

TABLE 2
Lysholm Knee Score 17 Years Postoperatively^a

	STG (n = 24) No. (%)	PT (n = 24) No. (%)	P Value
Excellent (95-100)	14 (58)	13 (57)	.616
Good (84-94)	8 (33)	6 (26)	
Fair (65-83)	2 (8)	4 (17)	
Poor (<65)	0	0	

^aPT, patellar tendon; STG, semitendinosus and gracilis tendon.

test comparing the involved limb with the contralateral limb. At 17 years, 20 patients (83%) in the STG group and 15 patients (63%) in the PT group had achieved a grade A hop, which represents a distance equal to or greater than 90% of that achieved with the contralateral limb ($P = .104$).

We found no difference in clinical instability between the two groups as determined using Lachman and pivot-shift tests ($P \geq .999$ and $P = .73$). In contrast, the STG group had a significantly higher number of patients with increased instrumented AP translation when using the KT-1000 arthrometer ($P = .03$) (Table 3). The mean value of anterior laxity measured with the KT-1000 arthrometer at manual maximum tension (side-to-side difference) was 2.17 ± 1.86 mm in the STG group compared with 1.33 ± 1.93 mm in the PT group ($P = .134$).

The AP translation measured in millimeters compared with the contralateral side and graded according to the IKDC during the follow-up period is shown in Figure 4.

We found no difference in clinically relevant joint crepitations between the two groups. Anterior knee pain was similar in both groups ($P = .376$). The differences were not significant and were not connected with patients' activities (walking, kneeling, climbing stairs). Kneeling pain was quite common, found in 46% of patients in the STG

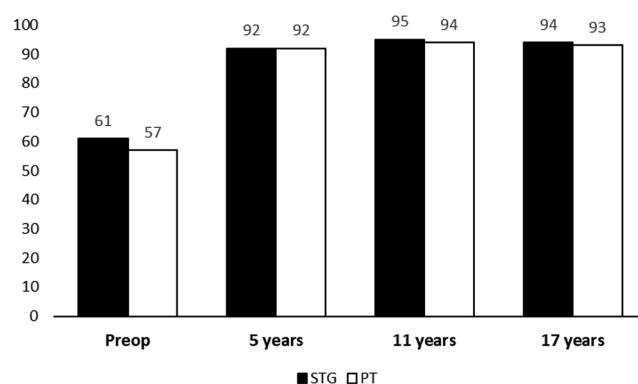


Figure 3. Average Lysholm score during follow-up period. PT, patellar tendon; STG, semitendinosus and gracilis tendon.

group and 33% of patients in the PT group ($P = .376$). Two patients in the PT group had minimal extension deficit compared with 1 patient in the STG group ($P = .551$).

Significant differences were found between the two groups regarding knee OA ($P = .04$). Seventeen years after ACL reconstruction, all patients in the PT group had radiologically apparent degenerative changes. In the STG group, 7 patients (29%) had grade A degenerative changes, according to the IKDC grading system. Eight patients (33%) in the PT group had grade C or higher degenerative joint disease compared with 5 patients (21%) in the STG group (Table 4). The differences were mostly in the patellofemoral and medial compartments ($P = .003$ and $P = .037$, respectively). We noted more degenerative changes of the lateral compartment in the PT group, although this was statistically insignificant ($P = .117$). The IKDC grading of degenerative joint disease during the follow-up period is shown in Figure 5.

The IKDC score was based on the original IKDC evaluation method and was not changed during the study

TABLE 3
Instability Examination 17 Years Postoperatively^a

	STG (n = 24) No. (%)	PT (n = 24) No. (%)	P Value
Manual Lachman test			≥.999
A (negative)	19 (79)	19 (79)	
B (positive with firm endpoint: 1+)	5 (21)	5 (21)	
C (soft endpoint: 2+)	0 (0)	0 (0)	
Manual pivot-shift test			.73
A (negative)	19 (79)	18 (75)	
B (glide: 1+)	5 (21)	6 (25)	
C (clunk: 2+)	0 (0)	0 (0)	
IKDC instrumented anteroposterior translation (KT-1000)			.03
A (<3 mm)	16 (67)	22 (92)	
B (3-5 mm)	8 (33)	2 (8)	
C (6-10 mm)	0 (0)	0 (0)	

^aIKDC, International Knee Documentation Committee; PT, patellar tendon; STG, semitendinosus and gracilis tendon.

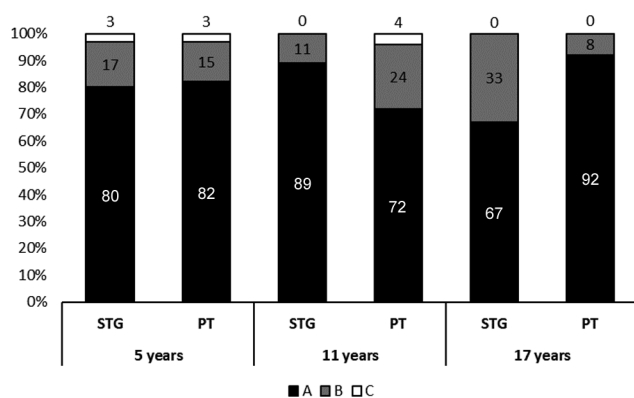


Figure 4. International Knee Documentation Committee grading of instrumented anteroposterior translation. PT, patellar tendon; STG, semitendinosus and gracilis tendon.

period. Table 5 represents the distribution of patients according to overall IKDC score. Overall IKDC evaluation grade A was obtained in 42% of patients in the STG group and 38% of patients in the PT group ($P = .82$) (Figure 6).

No differences were found between groups in health perception and health-related quality of life. Groups were comparable in all parameters of the Short Form-36 version 2 questionnaire (Figure 7). Normalized Physical Component Summary (PCS) was 51.6 in the STG group compared with 48.2 in the PT group ($P = .27$). Normalized Mental Component Summary (MCS) was 50.1 in the STG group compared with 49.9 in the PT group ($P = .95$). The PCS and MCS scores were lower compared with 11-year follow-up (PCS, 52.5 compared with 55.9; MCS, 52.1 compared with 50.3 respectively), especially PCS in the PT group.

DISCUSSION

Numerous studies have compared outcomes of arthroscopic ACL reconstruction with HT and PT grafts. Most of the

TABLE 4
International Knee Documentation Committee (IKDC)
Grading of Degenerative Joint Disease^a

Knee Joint Arthritis Grade	STG (n = 24) No. (%)	PT (n = 24) No. (%)	P Value
A	7 (29)	0 (0)	.04
B	12 (50)	16 (67)	
C	3 (13)	5 (21)	
D	2 (8)	3 (12)	

^aPT, patellar tendon; STG, semitendinosus and gracilis tendon.

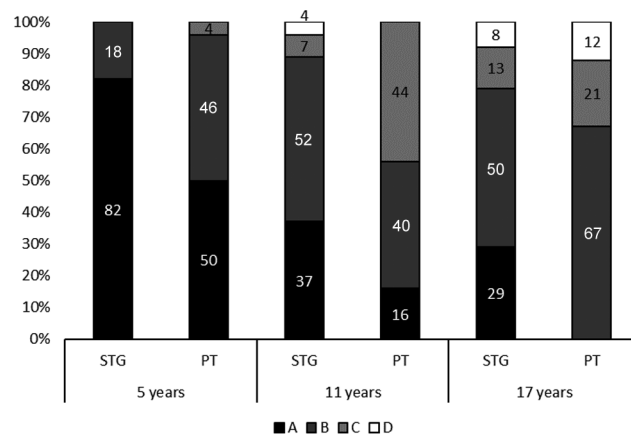


Figure 5. International Knee Documentation Committee grading of degenerative joint disease. PT, patellar tendon; STG, semitendinosus and gracilis tendon.

studies report short-term or intermediate outcomes with follow-up of less than 10 years.^{10,13,14,23,35} The aim of this study was to report outcomes at 17-year follow-up after ACL reconstruction with use of central-third bone-PT-bone autograft compared with double STG autograft. Apart from the graft, all other factors important for clinical outcome, such as fixation technique and rehabilitation

TABLE 5
Overall IKDC Score^a

IKDC Score	STG (n = 24) No. (%)	PT (n = 24) No. (%)	P Value
A (normal)	10 (42)	9 (38)	.82
B (nearly normal)	13 (54)	13 (54)	
C (abnormal)	1 (4)	2 (8)	
D (severely abnormal)	0 (0)	0 (0)	

^aIKDC, International Knee Documentation Committee; PT, patellar tendon; STG, semitendinosus and gracilis tendon.

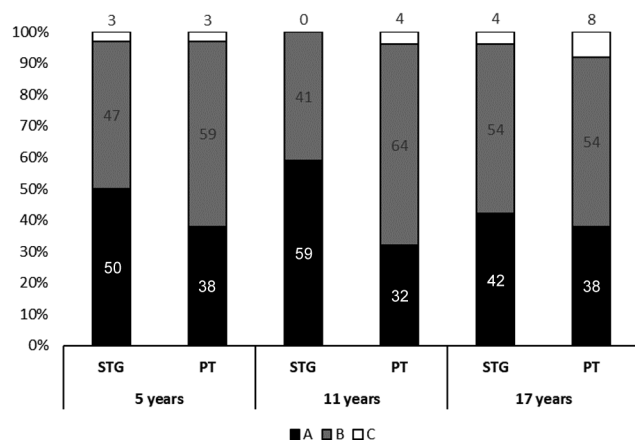


Figure 6. Overall International Knee Documentation Committee score during follow-up period. PT, patellar tendon; STG, semitendinosus and gracilis tendon.

protocol, were identical. One surgeon performed all of the reconstructions, and unbiased observers not involved in the operation or rehabilitation performed postoperative evaluations and arthrometric assessments. To our knowledge this is one of the longest follow-up periods among prospective RCTs comparing HT and PT autografts. Therefore, it is an excellent source of long-term outcomes after ACL reconstruction.

The main finding in the present long-term study is the increased prevalence of knee OA in patients with PT autograft. All patients in the PT group (100%) and 71% of patients in the STG group had at least some OA as revealed by radiography. As well, more patients in the PT group had high grades of degenerative changes compared with the STG group (33% of patients in the PT group had severe radiologic OA of grade C or higher, compared with 21% in the STG group). The differences were mostly in the medial and patellofemoral compartment. It is interesting that the degenerative changes were mostly asymptomatic and were mostly limited to the patellofemoral compartment. Thompson et al³² found knee OA in 61% of patients in the PT group and 41% of patients in the HT group, respectively ($P = .008$). Other authors reported similar findings, with higher prevalence of OA in the PT group.^{14,15,26,38} In contrast, some studies found increased

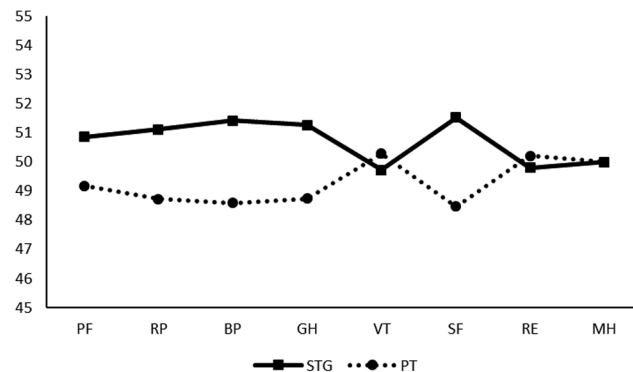


Figure 7. Mean Short Form-36 version 2 normalized sub-scores at the 17-year follow-up. Corresponding P values: PF (Physical Functioning, $P = .569$), RP (Role Physical, $P = .429$), BP (Bodily Pain, $P = .333$), GH (General Health, $P = .388$), VT (Vitality, $P = .847$), SF (Social Functioning, $P = .295$), RE (Role Emotional, $P = .894$), MH (Mental Health, $P \geq .999$). PT, patellar tendon; STG, semitendinosus and gracilis tendon.

prevalence of OA on postoperative radiographs compared with preinjury radiographs but found no difference between the HT and PT grafts.^{2,3,13,34}

The prevalence of OA is generally higher in our study because we included patients with concomitant chondral and/or meniscal injury. The PT group had a higher incidence of OA even though the preoperative and intraoperative findings were similar. The rates of meniscal surgery and chondrolysis found at time of surgery were not significantly different between the two groups ($P = .63$ and $P \geq .999$, respectively).

Chondrolysis is strongly correlated with knee OA. Patients in whom chondrolysis was found at the time of surgery had a 7.3 times higher odds of developing degenerative joint disease at 17-year follow-up compared with patients without chondral damage ($P = .035$). We also found a statistically significant correlation between subtotal meniscectomy and knee OA at 17 years in the STG group (odds ratio = 11.2, $P = .05$), in contrast to the PT group, in which no significant correlation was found (odds ratio = 1.3, $P = .76$). That can be explained by the fact that all of the patients in the PT group had at least some radiological changes and the effect of meniscectomy was lower.

No significant difference in graft survival was found between the two groups, with a 6.4% rate of ACL graft ruptures in the STG group and 10.3% in the PT group ($P = .636$). Our data also showed no difference in contralateral ACL injury; 9% of patients in each group had contralateral ACL rupture in 17 years. In contrast, Thompson et al³² found more graft ruptures in the HT group (18%) compared with the PT group (10%), although the difference was not significant ($P = .13$). They also found a significant difference in contralateral ACL injury between the two grafts—30% with PT graft and 14% with HT graft ($P = .035$). Those investigators attributed this finding to neglect of the contralateral knee during rehabilitation and the higher strain placed on the contralateral leg. Similar

findings were reported in a study by Webster et al.³⁴ We found no differences in contralateral ACL injury in our study, so we cannot confirm these findings. In a systematic review, Wright et al.³⁶ reported an incidence of ipsilateral and contralateral ACL injury that was similar to our findings.

Our study showed no differences in clinical instability between the two groups, although the STG group had significantly more patients with increased instrumented AP translation (33% compared with 8%, respectively). It is interesting that the average side-to-side difference increased in the STG group and decreased in the PT group compared with the results at 11-year follow-up (from 1.5 mm to 2.2 mm and from 2.5 mm to 1.3 mm, respectively).²⁷ No difference in manual pivot shift was found between the two groups: 21% of patients in the STG group had grade 1 positive pivot shift, compared with 25% in the PT group ($P = .73$). The proportion of positive pivot-shift test was higher in the STG group (7%) and lower in the PT group (28%) compared with 11-year follow-up. The reason for decreased AP translation and decreased number of patients with positive pivot shift in the PT group is probably attributable to greater progression of OA compared with the STG group.⁵ The results of a systematic review by Poehling-Monaghan et al.²⁵ are similar to our data, as those investigators did not find any differences in postoperative clinical knee instability. However, they also found no difference in instrumented AP translation. Other studies had similar results.^{32,34}

Our study showed no statistical difference in anterior knee pain and kneeling pain. It is interesting that patients had more kneeling pain in the STG group compared with the PT group (46% compared with 33%, $P = .376$) at 17-year follow-up. This is contrary to the results at 11-year follow-up, where 30% of the STG group and 48% of the PT group had kneeling pain. Other authors^{13,15,35} reported significantly more kneeling pain in the PT group with follow-up periods ranging from 9 to 15 years.

It is interesting that 46% of our patients were still at their preinjury level of activity at the 17-year follow-up. We found no differences between the two grafts in regard to patient activity measured with IKDC activity level and Tegner score. At 17 years after ACL reconstruction, 23% of patients were still participating in competitive sports. The average Tegner score at 17-year follow-up was lower by 1.75 compared with the preinjury score ($P = .004$). Other authors^{10,13,32} found similar decreases of activity level with no differences between the two groups.

We found no differences between the groups in terms of health-related quality of life, measured by the Short Form-36 version 2 questionnaire. The PCS and the MCS were lower compared with scores reported at 11-year follow-up. The decreased scores are mainly due to degenerative changes, as proposed in a systematic review by Filbay et al.⁸ Barenius et al.¹ also did not find any differences between groups or significant differences compared with an age-matched healthy Swedish population.

Overall, we found no significant differences between the two grafts, and both showed good to excellent long-term results with more than half of patients still at their preinjury level of activity 17 years after ACL reconstruction.

The patients' subjective evaluation of knee stability was high, with an average Lysholm score of 93 in both groups. The score is practically the same compared with the values at 5-year and 11-year follow-ups.^{27,28} The main differences between the 5-year and 11-year follow-ups are increased prevalence of OA, especially in the PT group, and increased instrumented AP translation in the STG group.

CONCLUSION

The results of this 17-year follow-up confirm that there are no major long-term differences between HT and PT autografts regarding subjective outcomes, quality of life, and clinical instability. Graft survival is similar in both groups. The STG group showed significantly more instrumented AP translation. The PT group had significantly greater prevalence of knee OA.

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